

Offline Learning

Human Demonstration



Unfinished OK!

Non-Contact

DMP Learning

Contact Index

Contact

Twist-Direction Segmentation

S1

S2

...

S_n

Screw Primitive Fitting

$$\pi_n = \{s_n, q_n, h_n, \theta_n^{end}\}$$

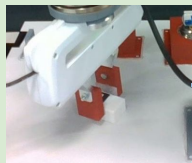
π_1

π_2

...

π_N

Forward Execution



DMP Motion



Contact Detect

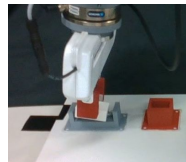
Geometric Consistent?

No

Yes

Spiral Search

Screw Primitives Execution



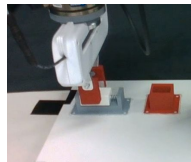
...

6D Admittance Pose Correction

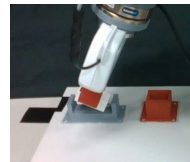
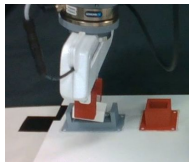
+

1D Admittance Speed Correction

Reverse Screw Primitives Execution



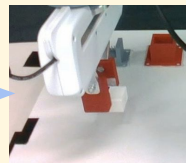
...



Contact Release Detection

Yes

Reversed DMP Motion



Edit Screw Parameters

$\tilde{\pi}_N$

...

$\tilde{\pi}_2$

$\tilde{\pi}_1$

Reverse Execution

No